

Lab 1

$R-2R$ Resistor Ladder (DAC)

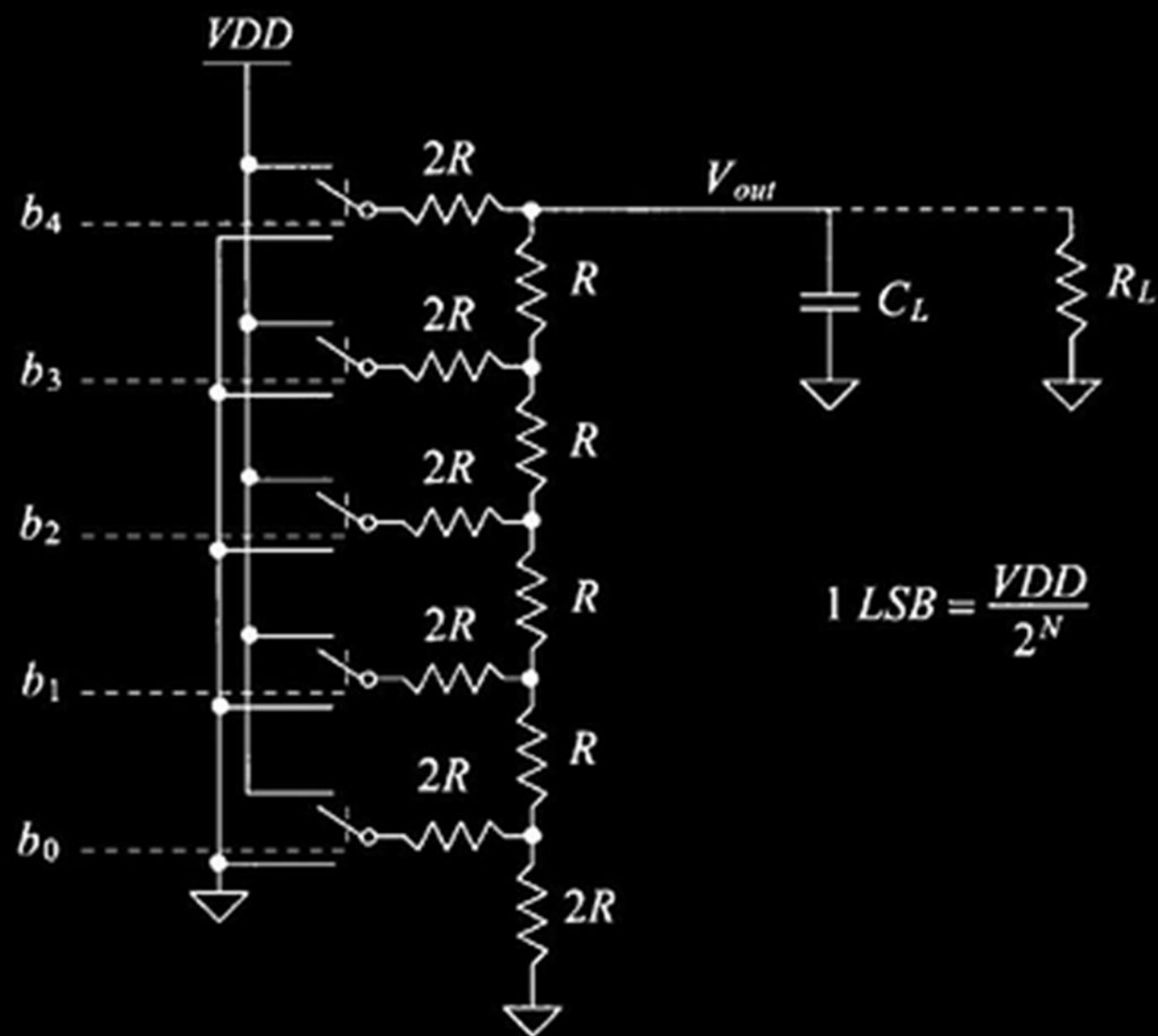
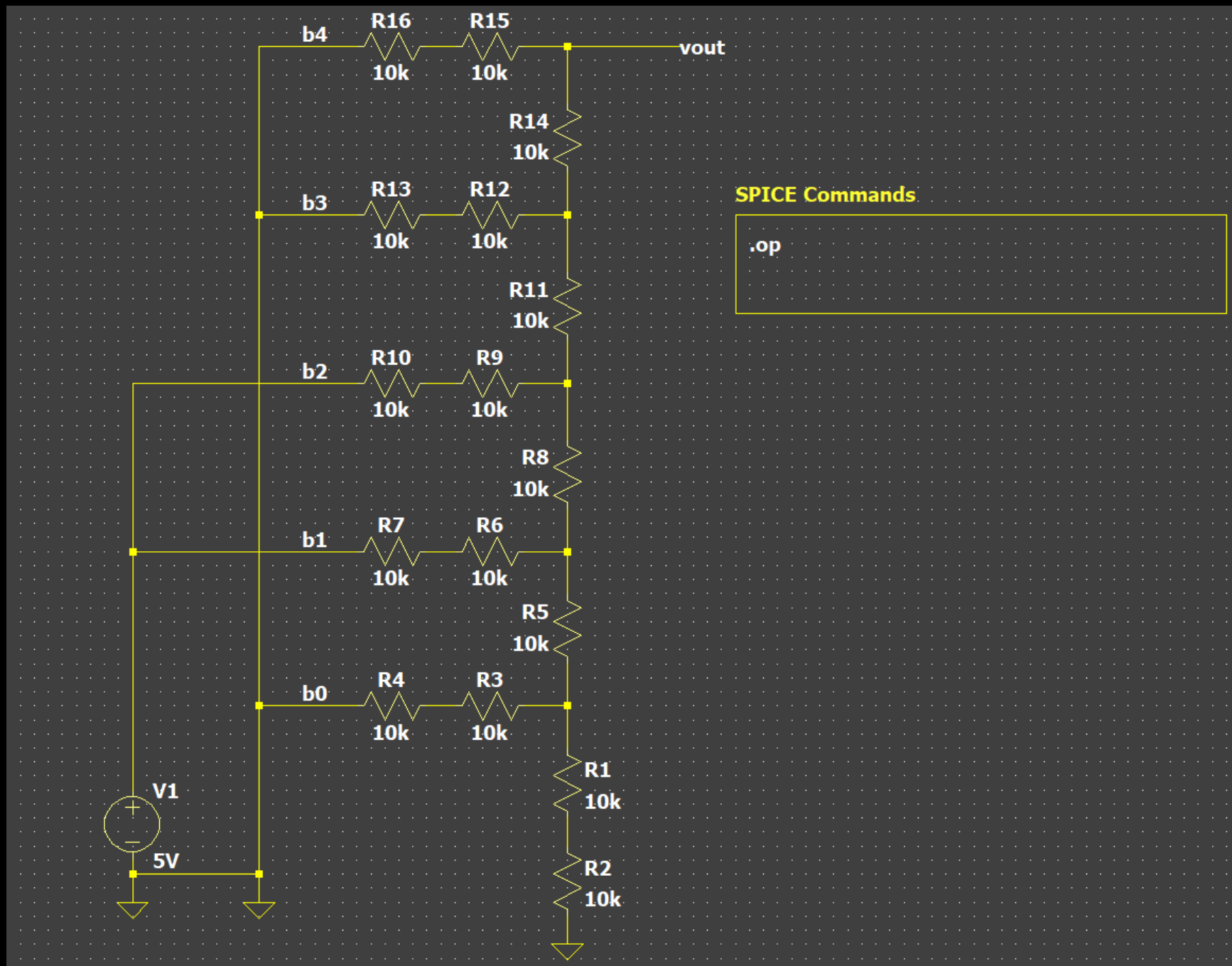
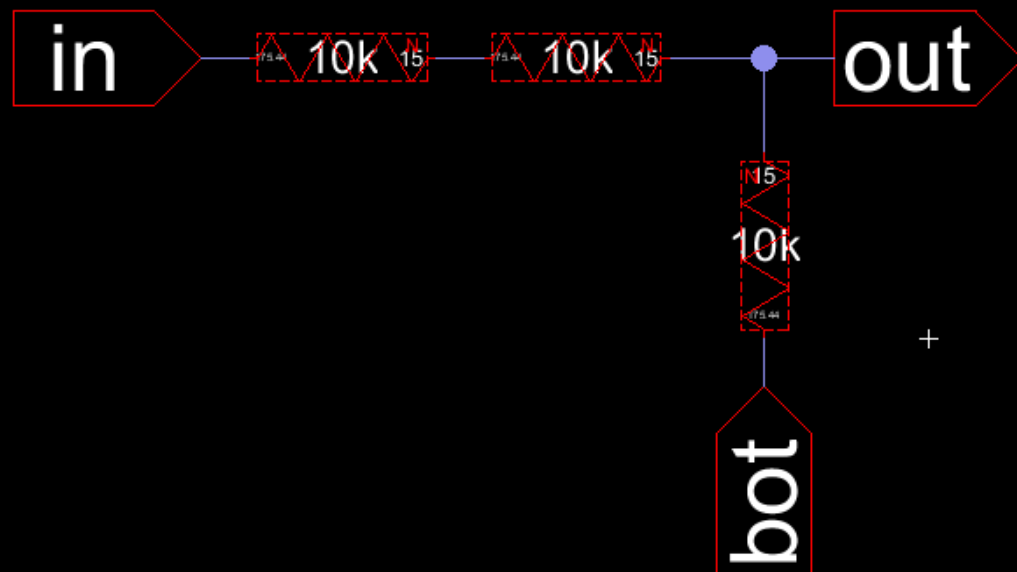
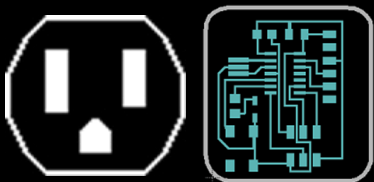
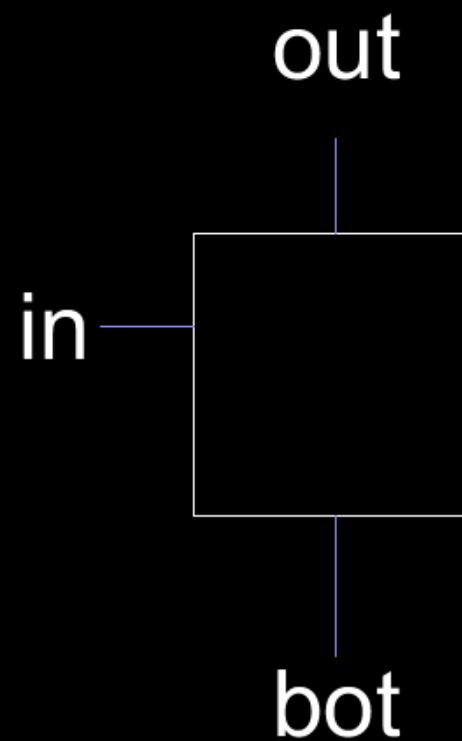


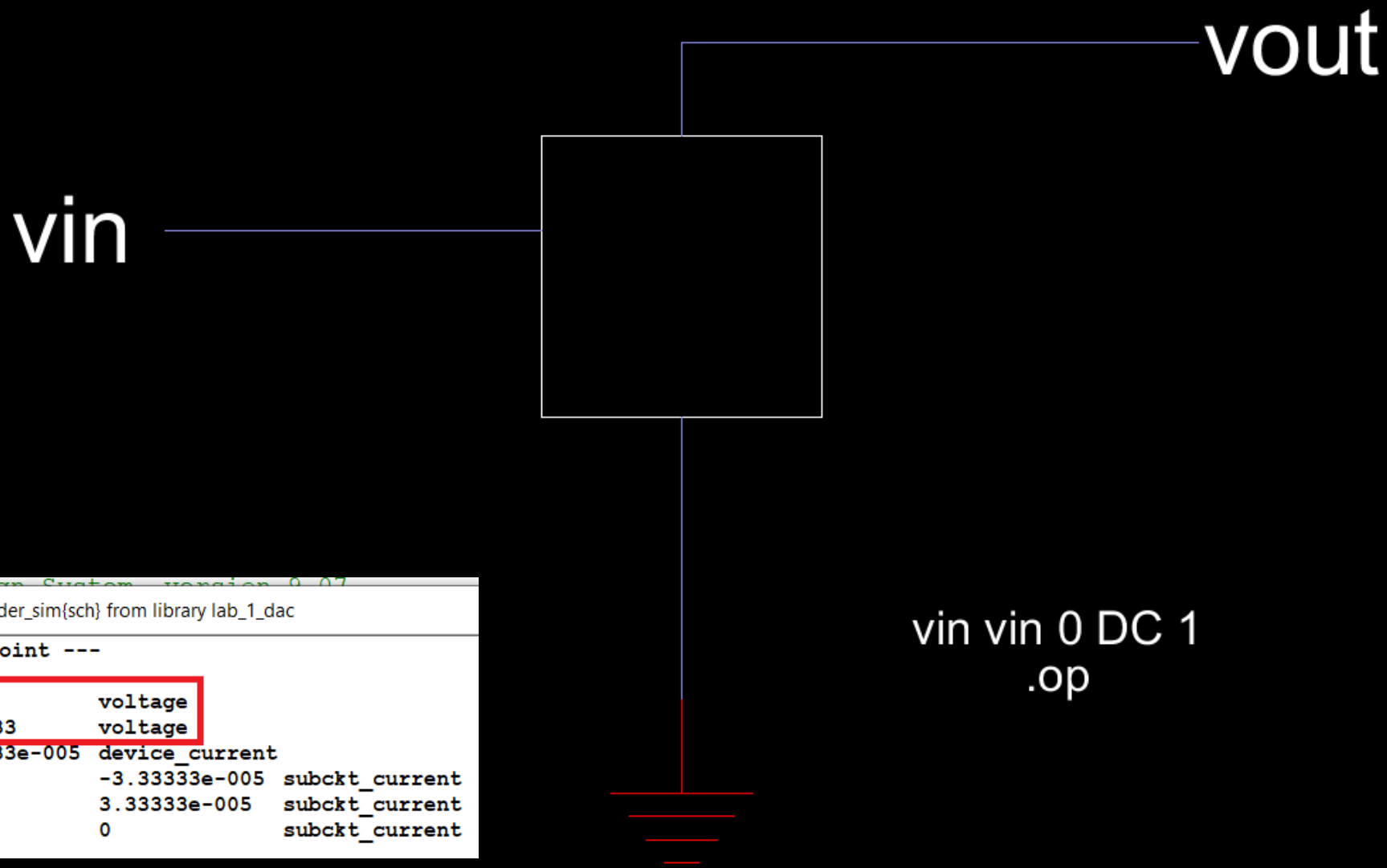
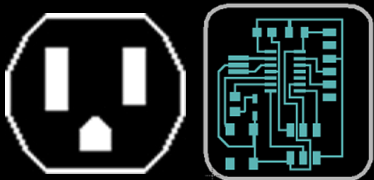
Figure 30.14 Voltage-mode (5-bit) DAC without an op-amp.





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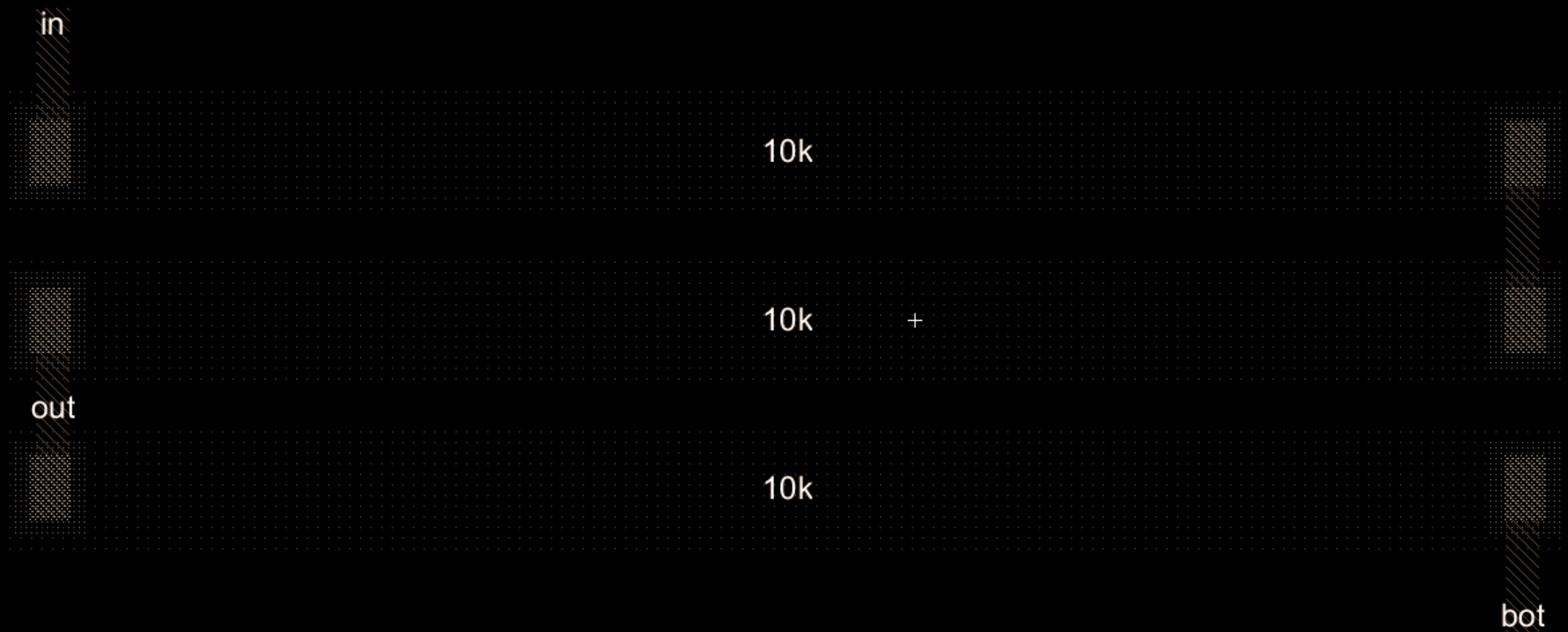
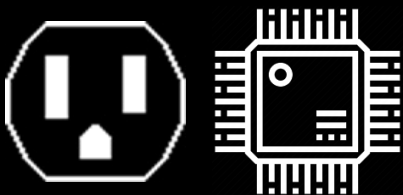


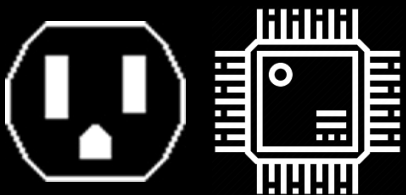
Electra VLSI Design System version 0.07

*** SPICE deck for cell R_divider_sim(sch) from library lab_1_dac

--- Operating Point ---

V(vin):	1	voltage
V(vout):	0.333333	voltage
I(Vin):	-3.33333e-005	device_current
Ix(r_divide@0:BOT):	-3.33333e-005	subckt_current
Ix(r_divide@0:IN):	3.33333e-005	subckt_current
Ix(r_divide@0:OUT):	0	subckt_current

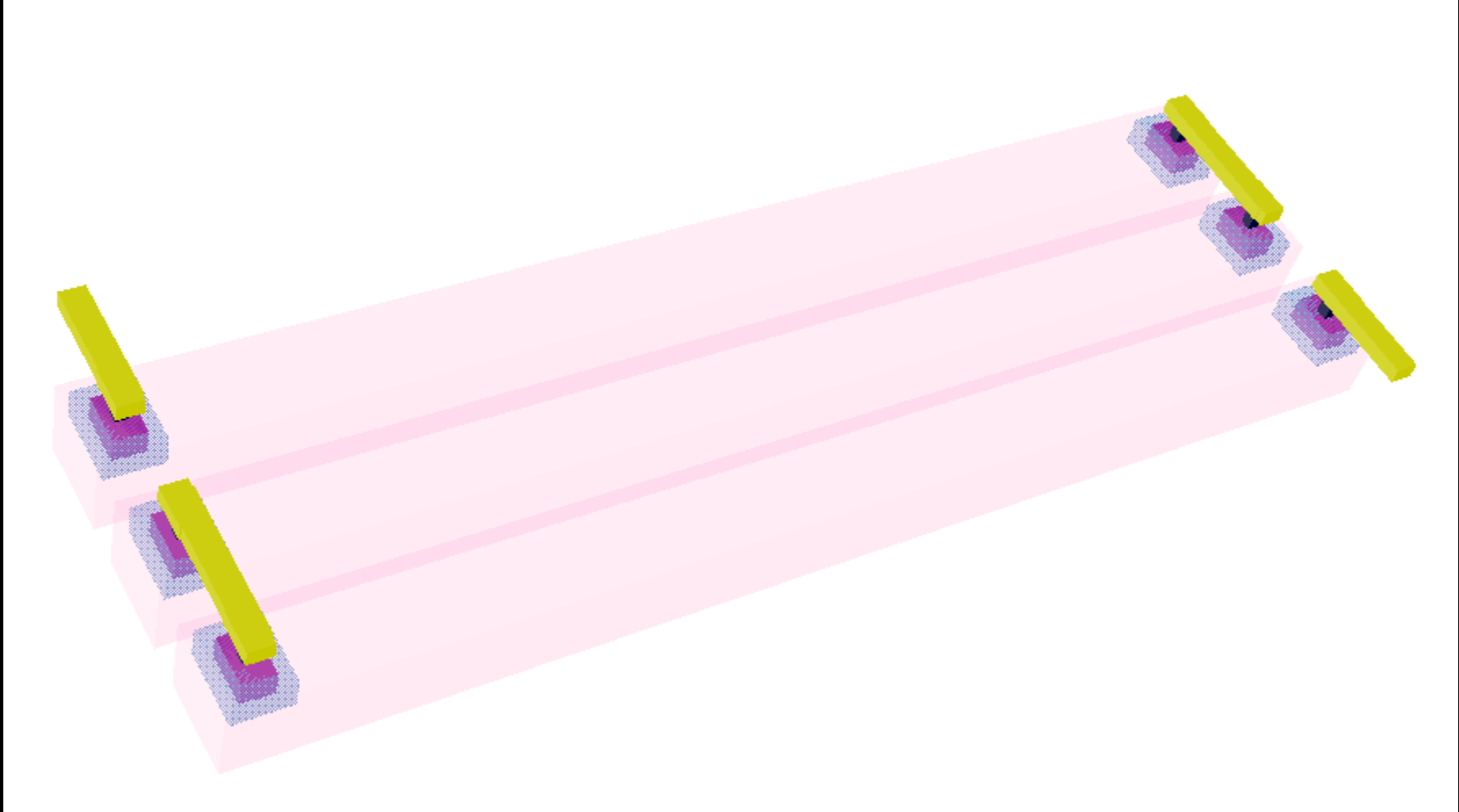
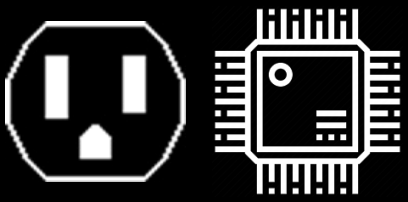


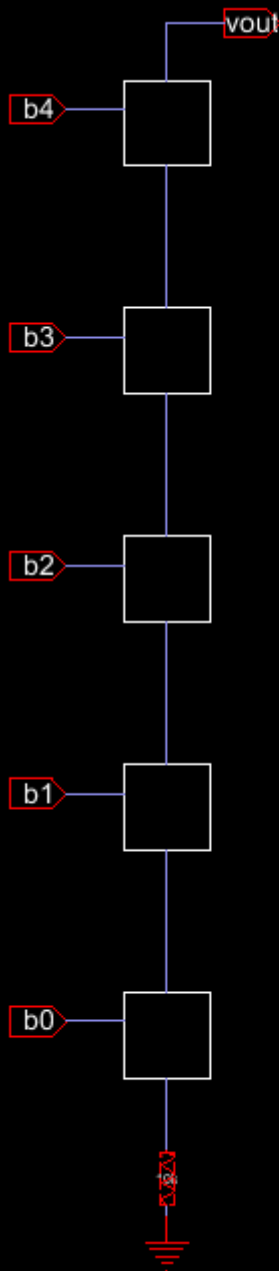
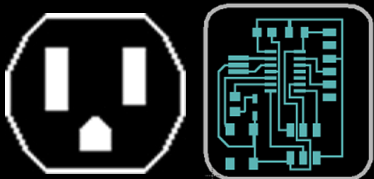


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Electronics Workbench Design System - version 6.0.0.7
*** SPICE deck for cell R_divider_sim{lay} from library lab_1_dac

--- Operating Point ---
V(vin) :      1      voltage
V(vout) :     0.333333 voltage
I (Vin) :     -3.33333e-005 device_current
Ix(r_divide@0:BOT) : -3.33333e-005 subckt_current
Ix(r_divide@0:IN) :  3.33333e-005 subckt_current
Ix(r_divide@0:OUT) : 0 subckt_current
```







```
+  
v4 b4 0  
v3 b3 0  
v2 b2 0  
v1 b1 b0  
vin b0 0 DC 5  
.op
```

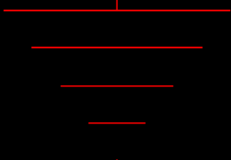
```
*** SPICE deck for cell 5Bit_DAC[sch] from library lab_1_dac  
  
--- Operating Point ---  
V(net@12):      1.25488      voltage  
V(b4):          0          voltage  
V(vout):        0.46875     voltage  
V(net@2):       0.703125    voltage  
V(b3):          0          voltage  
V(net@9):       1.28906     voltage  
V(b2):          0          voltage  
V(net@8):       2.51953     voltage  
V(b1):          5          voltage  
V(net@7):       2.50977     voltage  
V(b0):          5          voltage  
I(Rresnell@1):  0.000125488  device_current  
I(Vin):         -0.000248535 device_current  
I(V1):         -0.000124023 device_current  
I(V2):         6.44531e-005 device_current  
I(V3):         3.51562e-005 device_current  
I(V4):         2.34375e-005 device_current  
Ix(r_divide@0:BOT): 2.34375e-005 subckt current
```

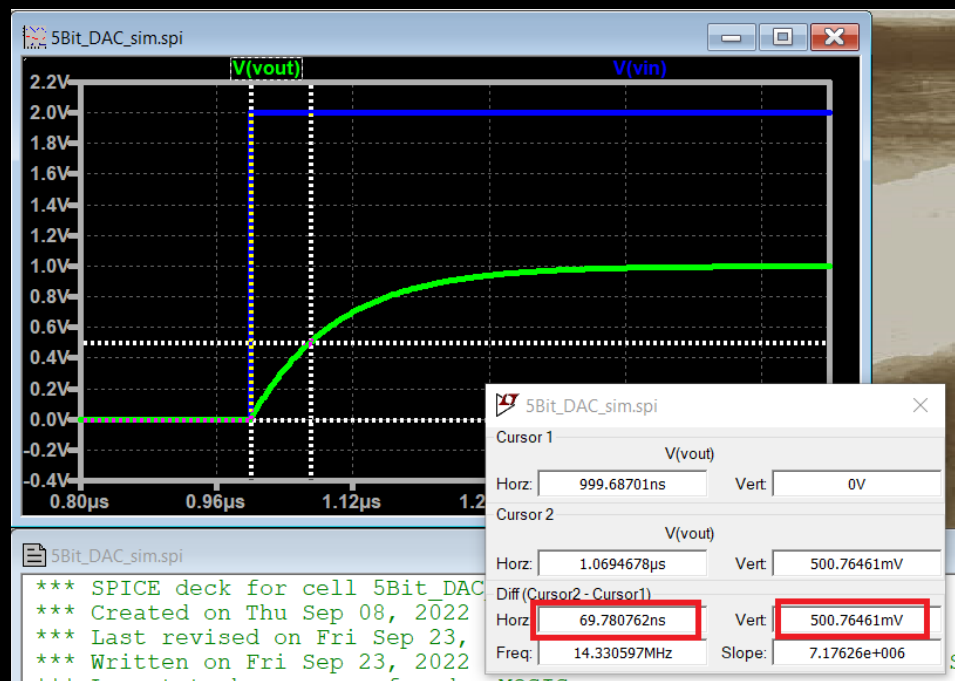
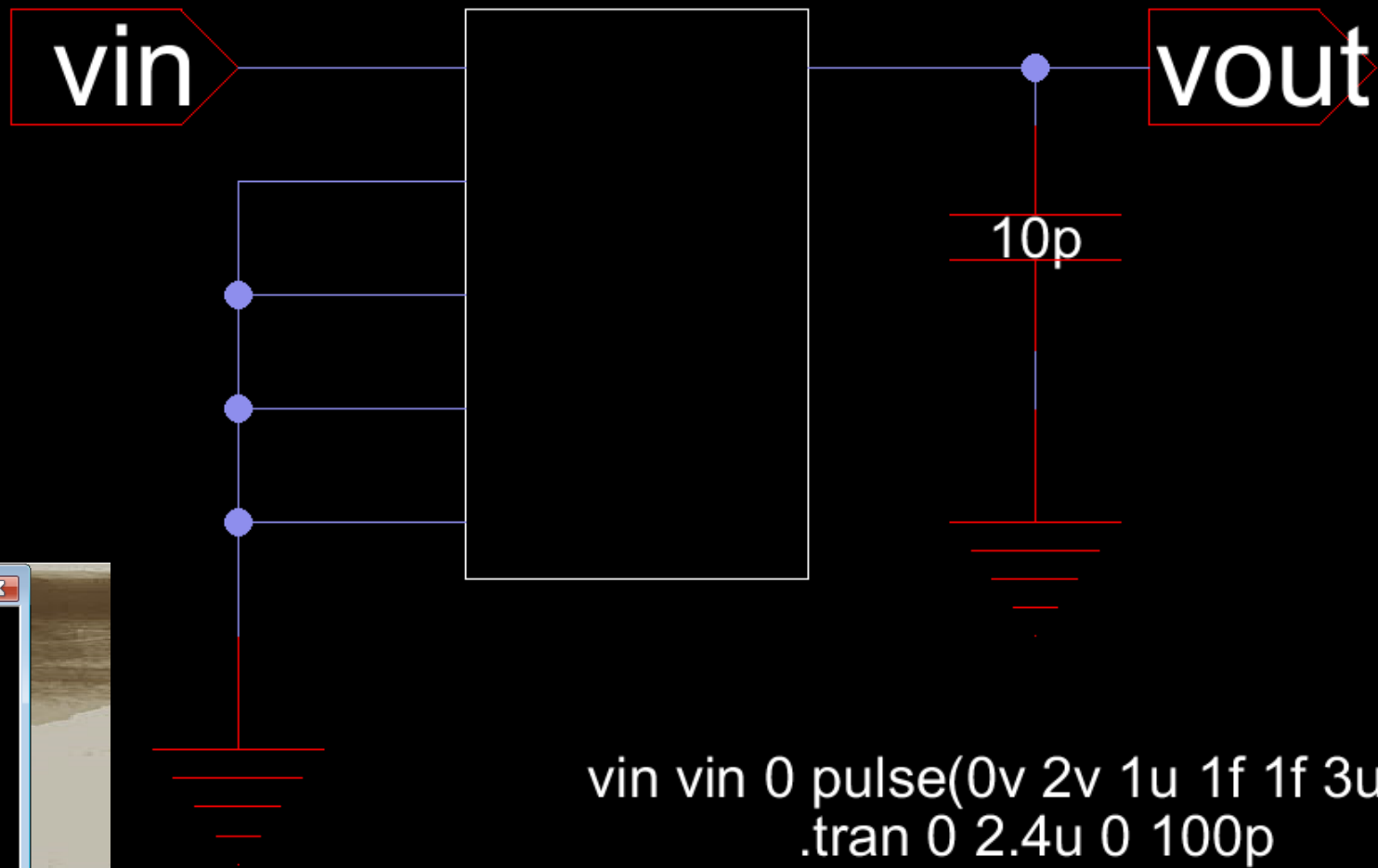
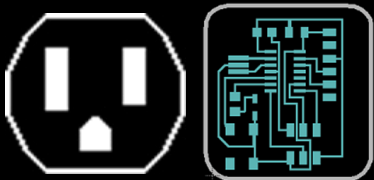


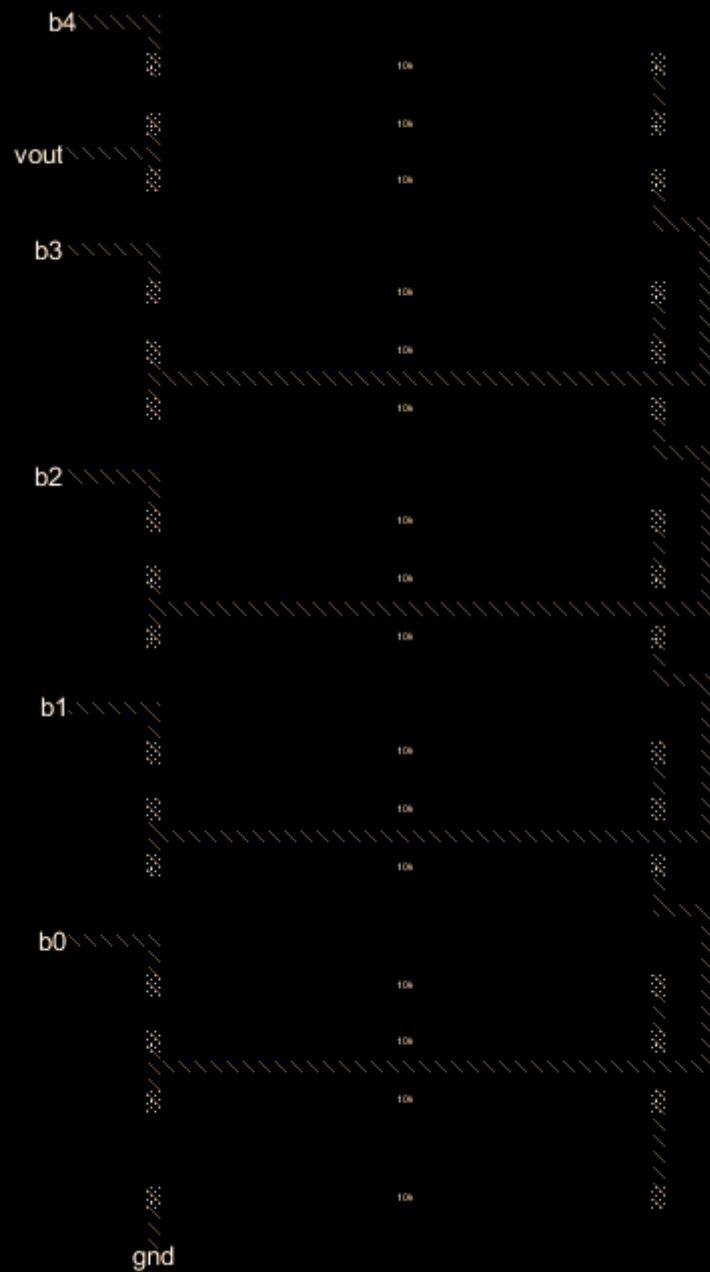
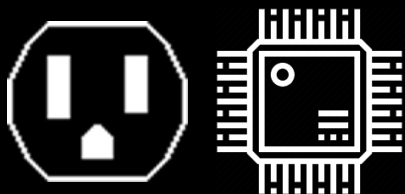
vin

vout

vin vin 0 DC 5
.op







+

v4 b4 0
v3 b3 0
v2 b2 0
v1 b1 0
vin b0 0 DC 5
.op

